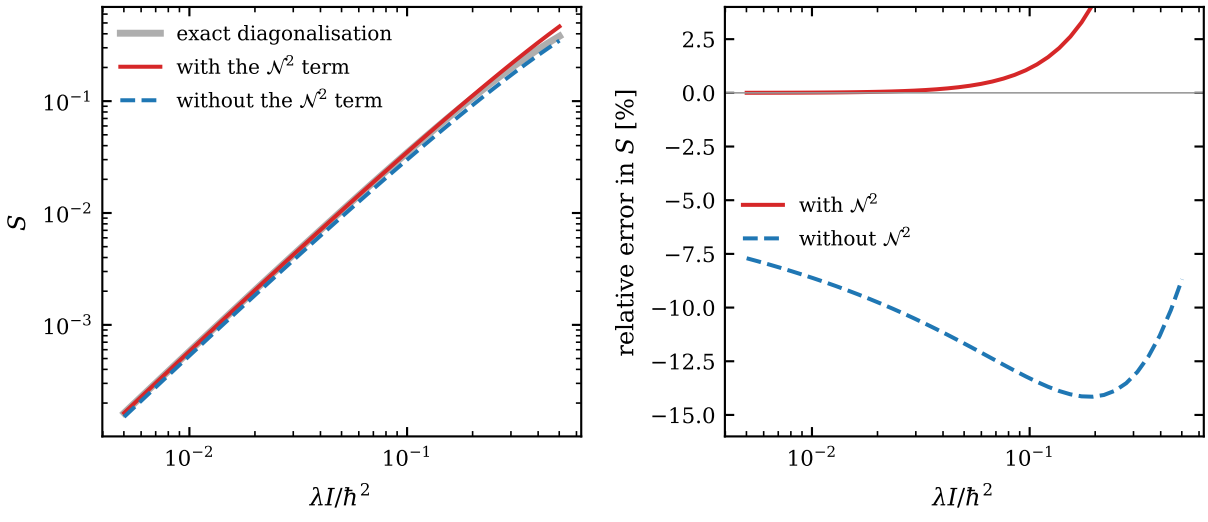
(a) weak coupling, $n = 0$
exact diagonalisation
with the $\mathcal{N}^2$ term
without the $\mathcal{N}^2$ term
S
10^{-1}
10^{-2}
10^{-3}
10^{-2}
10^{-1}
$\lambda I/\hbar^2$
(b) effect of the normalisation term
relative error in S [%]
2.5
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
−15.0
with $\mathcal{N}^2$
without $\mathcal{N}^2$
10^{-2}
10^{-1}
$\lambda I/\hbar^2$